Do Safety Constraints Tax Humor?

Quantifying the Alignment Tax via Distributional Analysis

Karthik Bhattaram

Humor Alignment Tax Project

May 29, 2026

Outline

The Core Question

Hypothesis

RLHF safety training compresses language model outputs toward safe, predictable text — reducing the statistical properties that make jokes *funny*.

Humor requires three things a safety-trained model might suppress:

- **Unexpectedness** — punchlines that violate expectations
- **Uncertainty** — keeping the listener guessing
- **Semantic distance** — punchlines that reframe the setup

We test this quantitatively across three models

Base $\longrightarrow$ Instruct (RLHF) $\longrightarrow$ Safety-finetuned (AdvBench + HarmBench)

Datasets

Joke corpora

- **r/Jokes** — 194,553 Reddit jokes (`title` + `body` format)
- **One-liners** — 3,200 short jokes (one joke per line)
- 1,000 jokes per source used for metric extraction

Safety fine-tuning data

- **AdvBench** — 520 harmful prompts
- **HarmBench** — 400 harmful prompts
- Total: 920 (prompt, refusal) pairs
- 7 varied refusal templates

Models

Model	Label	Training	Size
Llama 3.2-1B	Base	Pre-training only	1.2B params
Llama 3.2-1B-Instruct	Instruct	+ RLHF / instruction tuning	1.2B params
Llama 3.2-1B-Instruct-safety	Safety	+ LoRA refusal fine-tune	1.2B params

Safety fine-tuning: LoRA (r=16), 3 epochs, η=2$\times$10^{-4}, response-only loss masking on assistant turns. Loss: 3.72 $\rightarrow$ 0.34.

Three Distributional Metrics

1. Punchline Surprisal

Mean negative log-probability the model assigns to punchline tokens.
High surprisal $=$ model didn't predict the punchline.

2. Pre-punchline Entropy

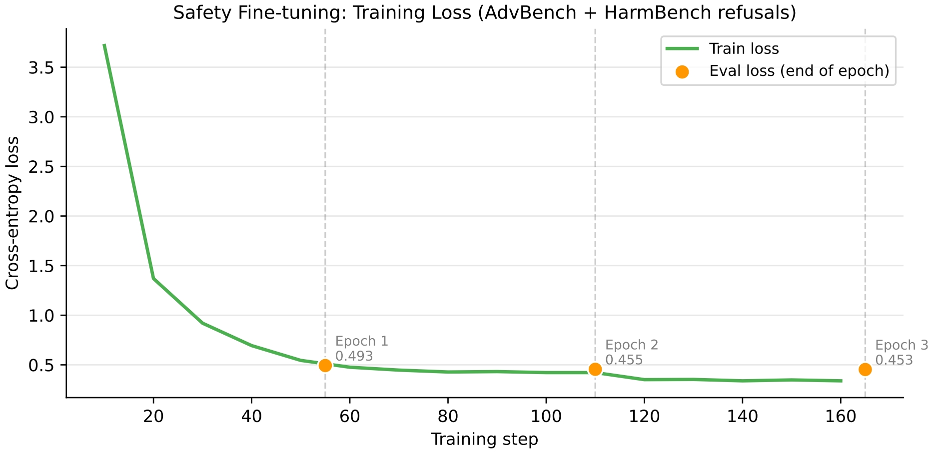
Shannon entropy of the next-token distribution at the end of the setup.
High entropy $=$ model uncertain about what comes next (many plausible endings).

3. Setup–Punchline Embedding Distance

Cosine distance between sentence embeddings of setup and punchline.
High distance $=$ punchline conceptually far from setup (semantic leap).

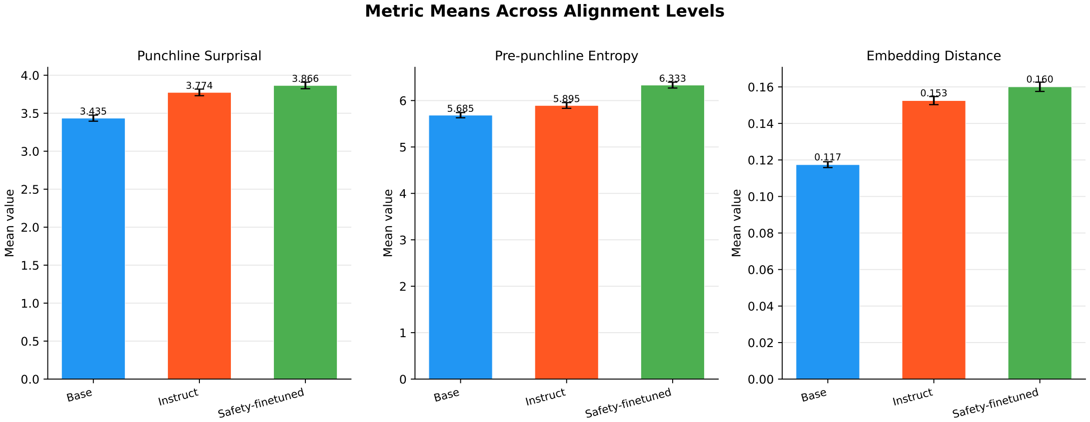
Inverted-U hypothesis: humor quality peaks at *moderate* values of all three.

Safety Fine-tuning: Training Curve

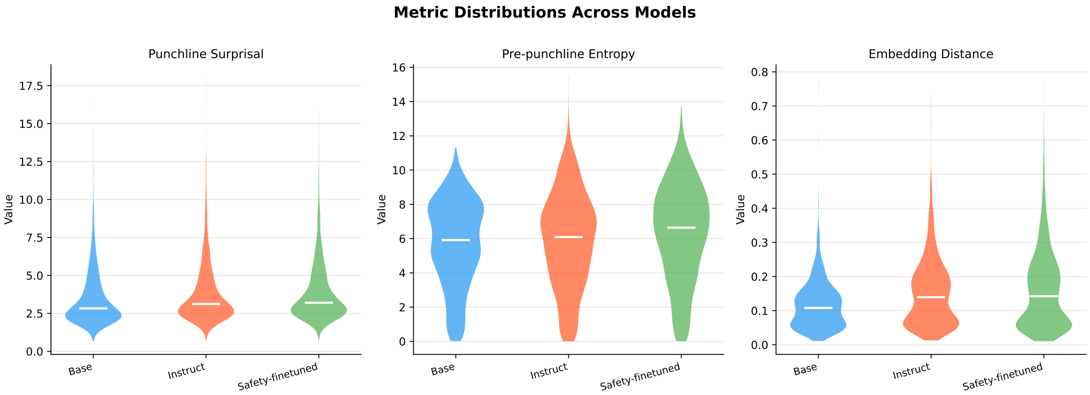


Loss converges from 3.72 (random) to 0.34 by epoch 3. Token accuracy on refusal responses reaches ≈ 90%.

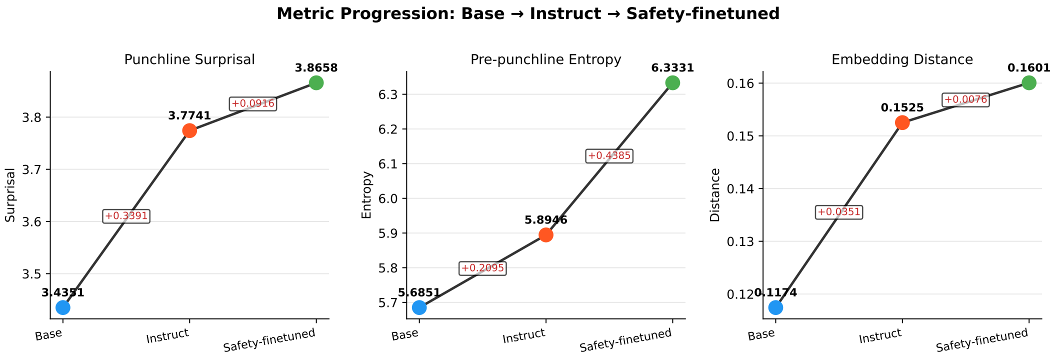
Results: Metric Means



Results: Full Distributions



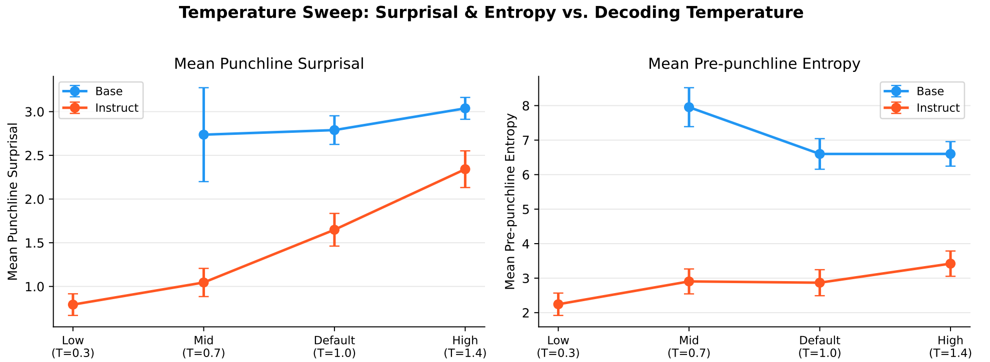
Results: Metric Progression



Results: Three-way Comparison (Statistics)

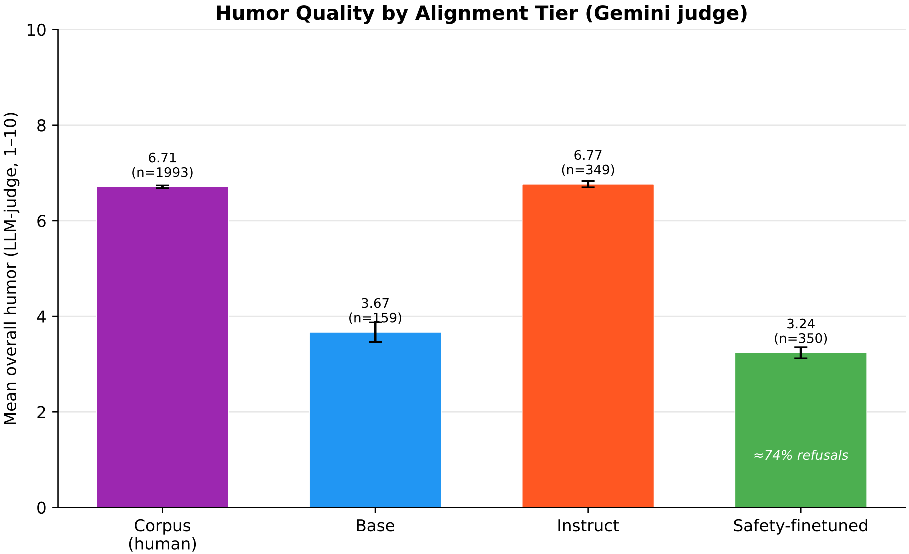
Comparison	**Metric**	**A mean**	**B mean**	**Shift**	**Cohen's** d	p
Base $\rightarrow$ Instruct	Surprisal	3.435	3.774	$+0.339$	$+0.18$	2×10^{-15} ***
	Entropy	5.685	5.895	$+0.210$	$+0.08$	7×10^{-2}
	Distance	0.117	0.153	$+0.035$	$+0.41$	2×10^{-27} ***
Instruct $\rightarrow$ Safety	Surprisal	3.774	3.866	$+0.092$	$+0.05$	1.6×10^{-2} *
	Entropy	5.895	6.333	$+0.439$	$+0.16$	1.1×10^{-7} ***
	Distance	0.153	0.160	$+0.008$	$+0.07$	0.56
Base $\rightarrow$ Safety	Surprisal	3.435	3.866	$+0.431$	$+0.23$	3×10^{-24} ***
	Entropy	5.685	6.333	$+0.648$	$+0.24$	5×10^{-14} ***
	Distance	0.117	0.160	$+0.043$	$+0.45$	4×10^{-25} ***

Results: Temperature Sweep



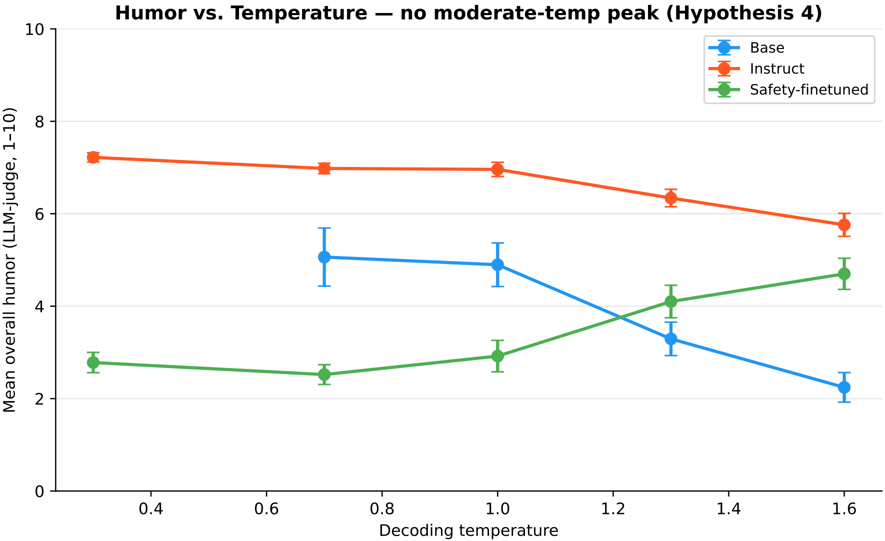
Higher temperature consistently raises both surprisal and entropy for both models. The instruct model remains above base at every temperature.

Humor Quality by Alignment Tier



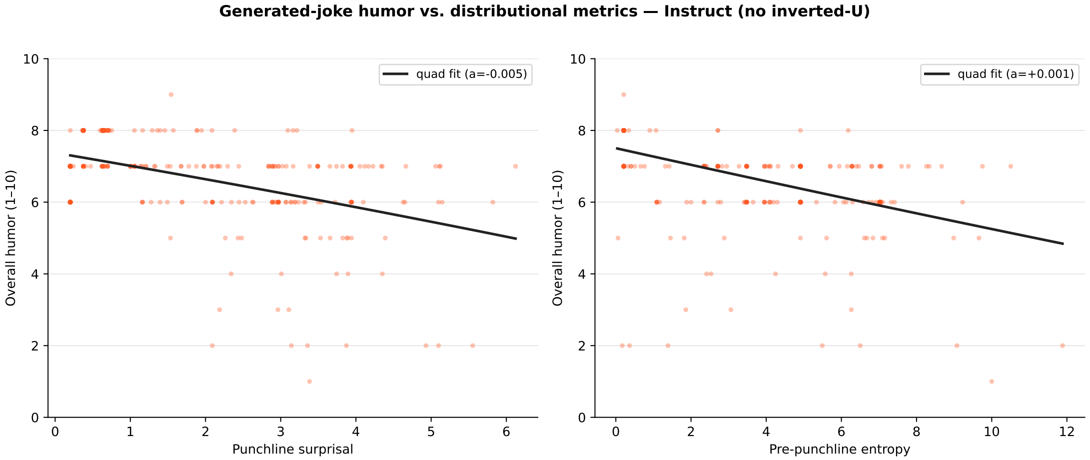
Gemini judge over 2,851 jokes (the local 1B judge produced valid scores for only ≈18% of generated jokes. Gemini

Humor vs. Temperature: No Moderate-Temp Peak



Hypothesis 4 not supported. As temperature rises, surprisal and entropy climb but instruct-model humor *declines*

Inverted-U on Generated Jokes? No.



Humor vs. distributional metrics for the instruct tier, with quadratic fits. No significant negative quadratic (no peak); metrics relate to humor more strongly than in the corpus (R^2 up to 0.32) but monotonically, not as a sweet spot.

Key Finding: Disruption, Not Compression

Opposite of the original hypothesis

All three metrics **increase** at every alignment step. Safety training does *not* compress outputs toward predictable text.

Two different mechanisms at work:

Base → Instruct
Driven by **surprisal** & **distance**.
RLHF makes the model genuinely surprised by punchlines — especially offensive ones it was steered away from.

Instruct → Safety
Driven by **entropy**.
Refusal fine-tuning disrupts the model's next-token confidence on *all* text (collateral damage from a narrow training distribution).

Content sensitivity: the instruct model finds offensive punchlines most surprising; clean jokes like "to get to the udder side" it predicts well.

Example Jokes: Largest Divergences

Instruct $\gg$ Base surprisal (offensive jokes):

- *What do you call a guy who gets lots of blowjobs?* $\rightarrow$ **Successful**
  Base: 14.2 nats, Instruct: 18.0 nats $(\Delta = +3.84)$
- *What can you make with epileptic lettuce?* $\rightarrow$ **A seizure salad**
  Base: 7.2 nats, Instruct: 10.7 nats $(\Delta = +3.58)$

Base $\gg$ Instruct surprisal (classic clean jokes):

- *Why did the hungry baby calf cross the road?* $\rightarrow$ **To get to the udder side.**
  Base: 7.1 nats, Instruct: 5.7 nats $(\Delta = -1.43)$
- *For Sale: French WWII Rifle* $\rightarrow$ **Never fired. Only dropped once.**
  Base: 7.7 nats, Instruct: 6.5 nats $(\Delta = -1.19)$

Interpretation: RLHF confers familiarity with classic joke formats and surprise at harmful content — not uniform compression.

Conclusion & Future Work

Findings:

- Inverted-U curve: not confirmed at 1B scale (noisy signal, low R^2)
- Alignment shift: confirmed — but direction is *upward*, not downward
- Two distinct mechanisms: RLHF content-sensitisation vs. safety fine-tuning entropy disruption

Limitations:

- 1B model is weak — effects may differ at 7B+
- Dataset contains many offensive jokes that confound the shift signal
- Refusal fine-tuning used only 920 examples over 3 epochs

Future work:

- Filter to clean-only jokes and re-test the compression hypothesis
- Scale to Llama 3.1-8B
- Add LLM-as-judge humor scores to test inverted-U at larger scale

Thank you!

Code: `github.com/karthikb/humor-alignment-tax`